



Sure, AI Can Automate. But How Can You Use It to Innovate?

A Kellogg professor's experience deploying AI in the classroom shows how domain knowledge and experimentation can lead to true breakthroughs.



BASED ON INSIGHTS FROM

Sébastien Martin

SUMMARY Using ChatGPT to automate tasks you already do may seem efficient, but the real way of unlocking AI's transformative potential is to use it to collaborate on innovations in the workplace. This requires a bottom-up approach to empo...
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When [Sébastien Martin](#) used ChatGPT to create an AI teaching assistant for his operations management course, [the initial reviews were good](#). Students used the chatbot, named “Kai,” to review lectures and slides, create practice quizzes, and take pre-class assessments.

But soon, Martin started to wonder if he wasn't being ambitious enough about how AI could transform education.

Merely automating the smaller, routine tasks of his TAs was nice—“AI doesn’t get tired,” he says—but not a true innovation.

“This is where I think most of the misunderstanding is around AI,” says Martin, associate professor of operations at Kellogg. “Often, people’s first instinct is to use AI to automate something they were already doing. This may not sound like a trap, but I think it is.”

He sees a common trend in how most AI users adopt the technology, moving from initial fear to tentative use for replacing routine tasks. But it’s the next two steps—

collaboration and innovation—where the transformative potential of AI really emerges.

Since those initial trial runs with Kai, Martin has now built new AI tools that facilitate collaboration with human TAs, personalize homework assignments, and power a novel type of interactive case study, one of which was recently featured in [The Wall Street Journal](#).

Martin’s experience offers lessons beyond the classroom for business leaders who are currently struggling with how to incorporate AI into their operations.

“It’s something fundamentally new,” Martin says. “The

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amazing power of this tool can be used to do completely new things we couldn't do before, and these things have insane value. Never has it been easier to innovate—to create true groundbreaking innovations, even at companies that have been around for decades.”

Promote bottom-up collaboration with AI

In the business world, there is no shortage of consulting firms and what Martin refers to as “LinkedIn gurus” offering comprehensive theories of the future of AI and promising to implement an AI strategy at your company.

Instead, he suggests, think about AI in terms of opportunities for your business. And who knows your business better than you and your employees? In today's uncertain environment, he notes, “there is no specific plan that you know will work for sure,” so you're better off crafting it yourself.

Letting specific teams come up with their own applications entails trusting them to notice where AI gets things wrong or where the human touch might be more appropriate.

“The people in your company are, by definition, the experts at what they're doing. They know

what's right; they know what's wrong," Martin says.

In Martin's class, he noticed that certain student questions required more assistance than the AI TA could provide. For example, a student who will miss a future class because of a wedding wants to know not just the material that will be covered but other tasks they will need to complete to make up for the absence. Martin rewrote Kai to direct these more-complex inquiries to his human TAs or himself—an adjustment that created more emails for his team to answer but also created higher-quality interactions with his students.

“What's worked really well is actually bridging it with my TAs as much as possible,” Martin says. “It's very hard to figure out exactly what you want to give to the AI and what you want to handle with humans. It's deeply personal. It's problem-specific, it's instructor-specific, and it requires much more domain knowledge. It's not just about plug-and-play but finding the right niche for the tool.”

Martin also found human-AI collaboration effective for allaying common concerns about how the technology fits into education. Many educators fret that students will now just use large language models to feed them homework answers instead of doing the work

themselves, while students complain of their instructors using the technology in grading. Martin looked at it from a different perspective—what if the conversational abilities of these models could be used to demonstrate students’ knowledge of the material?

Instead of asking his operations students to answer a list of quiz questions, Martin challenged them to “teach” Kai, which he instructed to role-play as a first-year MBA student. The human student explains a concept, such as supply-chain management, to the AI model, which responds with periodic questions about the topic. The student then submits the log of

that conversation to the human TA, who grades it based on accuracy and how seriously they took the exercise.

Students have responded enthusiastically to the new homework, Martin says, and it has the additional benefit of being cheat-proof. This process shows how flipping the script on a common fear related to AI—in this case, that it’s a magical “homework machine”—can lead to benefits.

For companies that are starting to move past the cautious experimentation phase with AI, he suggests empowering employees with the resources and space to play around with AI, get comfortable with it, and

figure out on their own where it best fits into their operations—and where it can be used to move beyond automation to innovation.

“The world has changed with AI,” he says. “Now there’s gold hidden everywhere. It’s a very rare situation where any employee anywhere can figure out something amazing in a matter of weeks.”

Find the truly new

The ability of AI to role-play, taking on different personas based on training data and instructions, inspired Martin’s next ambitious idea to reinvent his classwork. A staid tradition in business education is the

case study, a lengthy document that gathers real-world research—interviews, data, supporting documents—about a business or organization facing a challenge. Students are typically asked to read a report and answer questions on the most important lessons from the study.

Martin wondered if he could use the shape-shifting personality of AI to enrich this experience. Instead of merely reading the case study and regurgitating conclusions, students could conduct a virtual case study themselves.

To test this concept, Martin, together with professor [Daniela Hurtado-Lange](#), developed an

AI-powered case study about a school district facing a transportation crisis. The case study provides a virtual cast of stakeholders students can interview, each with their own perspectives and blind spots based on real interview data.

For example, students can talk to the superintendent, who is the main decision-maker and is particularly aware of the political landscape. Or they can talk to a veteran bus driver, who has on-the-ground knowledge and truly knows what happens on their bus routes.

The AI can also directly share data and documentation with students. For example, the COO

of the school district can provide the student with a transportation simulator and give them instruction in how to operate it.

“It’s much richer than a usual case in the sense that it’s interactive, you have different characters with different points of view, you can ask any follow-up question that you want, and so you have unreliable narrators, where you’re not sure whether you can trust everyone,” Martin says. “You can do roughly the same thing you can do with a traditional case study, but you can also do much more.”

Where the conclusions and recommendations in the

traditional case study were delivered by the experts who wrote it, students in the AI-powered version have to piece the puzzle together themselves and come to their own conclusions.

“This is something you couldn’t do before,” Martin says. “It’s open-ended, creative, and it feels much more like a real-world scenario.”

Martin was able to implement this innovative idea over a weekend of work (after months of thought). But even for a novice, figuring out how to get AI to do something useful is surprisingly simple. In his experience, exposure to LLMs, bit by bit, can make people

comfortable with the tools and their capabilities.

“Anybody—even those without a technical background—can, just by talking to ChatGPT regularly, see a huge difference in their understanding of this machine and how it works,” he says.

For companies, investing time into training employees will help people overcome their fear of the technology and, even more importantly, begin to feel some mastery over it. This can help them get excited about its possibilities.

“Once people know a little bit about AI, they become curious—that’s a very natural human

thing,” Martin says. “If you show people how to do something, they realize it’s actually not that crazy, and then they have all sorts of ideas of cool things to do. They find value in what they’re doing, so they’ll be excited about improving.”

Ultimately, he says, AI presents an extraordinary opportunity for firms to unlock creativity and new solutions—if they can get comfortable with the technology, ensure that its applications are coming from the bottom up, and stay nimble. Companies should be looking to AI less for its potential to facilitate workforce reduction than for its potential for

innovation and competitive edge.

“Putting this tool in the hands of the people in your company, and removing these obstacles, will, in the future, allow you to react faster,” Martin says. “So, if a new iteration of an AI technology comes out and everyone is freaking out, now you’re the company whose employees are not only not afraid of it, but have played around a little bit with it and know its strengths and weaknesses.”

FEATURED
FACULTY

Sébastien
Martin

Associate
Professor of

ABOUT
THE
WRITER

Anna
Louie
Sussman